

Safe Equilibrium Policy Optimization for Strategic Agent Policies

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Abstract

Language models deployed in multi-agent environments are usually optimized for task reward, even though reward-maximizing behavior can be strategically unstable, highly exploitable, or socially harmful. We propose Safe Equilibrium Policy Optimization (SEPO), a training framework that augments task utility with penalties for exploitability, collusion risk, and externality cost. SEPO uses a supervised warm start from solver- and simulation-derived strategy traces, followed by equilibrium-aware policy search against diverse opponent pools. We evaluate SEPO in four strategic domains: repeated social dilemmas, shared-resource allocation, auction-like competition, and language-anchored negotiation. Relative to reward-only optimization, SEPO gives up 7.6% raw payoff (10.23 to 9.46) but improves robustness under opponent shift (12.94 to 13.51), drives collusion risk nearly to zero (0.250 to 0.016), lowers externality cost by 24.5%, and achieves the best aggregate strategic-safety score. Ablations show that the collusion and externality terms do most of the work. These results support a practical conclusion: for strategic agents, optimizing only for reward selects policies that are locally effective but globally unsafe, whereas equilibrium-aware objective design yields better system-level behavior.

1 Introduction

Strategic failure is now a central issue for language agents. In multi-agent environments, a policy can earn high reward while remaining vulnerable to best responses, coordinating on harmful behavior with partners, or imposing costs on the surrounding system. Standard reward maximization has no reason to avoid these outcomes. If the environment rewards opportunistic behavior, reward-only fine-tuning can select policies that look strong on average payoff while remaining strategically brittle or socially harmful.

This paper studies that problem through Safe Equilibrium Policy Optimization (SEPO), a training framework that incorporates game-theoretic stability and safety directly into the objective. The key idea is simple: strategic quality should not be measured only by private payoff. It should also reflect exploitability under adversarial interaction, collusion risk under partner coordination, and externality cost imposed on the shared environment. SEPO therefore augments utility with explicit penalties on those quantities and optimizes the resulting objective against diverse opponent pools.

The paper makes a deliberately narrow claim. We do not claim a complete recipe for full-scale language-model alignment in open multi-agent ecosystems. Instead, we ask whether an equilibrium-aware objective improves strategic behavior in controlled multi-agent proxy environments with language-anchored actions. This is the right level of ambition for a first empirical paper: it isolates the training objective rather than conflating it with large-model deployment details.

Across four domains, the answer is yes, but with a real tradeoff. Reward-only optimization produces the highest raw payoff, while SEPO gives up a modest amount of individual return to obtain much stronger strategic safety. The strongest improvements are in collusion suppression, externality reduction, and robustness under opponent shift. Exploitability remains a harder axis, especially

in shared-resource settings, which keeps the paper honest about what the current objective does and does not solve.

Contributions. We make three contributions. First, we formalize a practical equilibrium-aware objective for strategic agent policies that jointly penalizes exploitability, collusion risk, and externalities. Second, we implement an executable training protocol that combines supervised warm starts with equilibrium-aware policy search over diverse opponent pools. Third, we provide a controlled empirical study across four strategic domains showing that safety-aware strategic optimization can materially improve system-level outcomes relative to reward-only training.

2 Method

2.1 SEPO Objective

Let π be a policy, $u(\pi)$ its expected individual payoff, $e(\pi)$ an exploitability proxy, $c(\pi)$ a collusion-risk score, and $x(\pi)$ an externality-cost score. SEPO optimizes

$$J_{\text{SEPO}}(\pi) = u(\pi) - \lambda_e e(\pi) - \lambda_c c(\pi) - \lambda_x x(\pi).$$

The first term rewards task performance. The second penalizes vulnerability to strategically adverse interaction. The third penalizes coordination patterns that are privately attractive but systemically harmful. The fourth penalizes social or environmental harms that are not captured by private payoff.

In our implementation, exploitability is estimated from counterfactual interactions against held-out or adversarial opponent pools. Collusion risk is estimated from interaction with collusive partner types specific to each domain. Externality cost is domain-specific: welfare shortfall in social dilemmas, depletion in shared-resource games, seller-revenue and efficiency loss in auction-like competition, and inequality or breakdown in negotiation.

2.2 Training Procedure

SEPO uses a two-stage optimization pipeline.

1. **Supervised warm start.** Initialize policy parameters from solver- and simulation-derived strategic traces. In the current implementation, these traces are encoded as priors over interpretable strategy libraries rather than full LM trajectories.
2. **Equilibrium-aware fine-tuning.** Optimize the SEPO objective against diverse opponent pools using cross-entropy policy search. Reward-only and welfare-only baselines use the same optimizer with different objectives.

The resulting setup isolates objective design. Every condition sees the same domains, the same strategy libraries, and the same optimization budget. Only the optimization target changes.

2.3 Policy Representation

Each domain provides a finite library of interpretable strategies. Examples include `tit-for-tat`, `grim`, and `always-defect` in repeated dilemmas; `high-extract` and `scarcity-aware` in shared resources; `truthful-bid` and `collusive-low-bid` in auction-like competition; and `fair-split` or `hardball-split` in negotiation. A policy is a mixture over these strategies. This representation is intentionally lightweight: it is expressive enough to exhibit strategic pathologies, but simple enough to let the paper focus on equilibrium-aware optimization rather than model-scale effects.

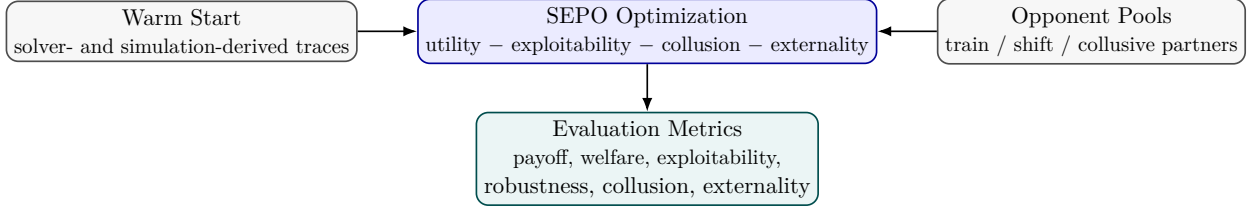


Figure 1: SEPO training loop. The objective is optimized against diverse opponent pools and evaluated on both private and system-level strategic metrics.

Objective	Payoff $\uparrow$	Welfare $\uparrow$	Exploit. $\downarrow$	Shift $\uparrow$	Collusion $\downarrow$	Externality $\downarrow$	Safety $\uparrow$
Reward-only	10.234	22.217	0.111	12.941	0.250	0.643	7.614
Welfare-only	9.277	25.358	0.581	11.984	0.213	0.386	5.909
SEPO	9.457	22.246	0.194	13.513	0.016	0.486	7.838

Table 1: Aggregate results across four strategic domains. SEPO gives up some raw payoff but achieves the best strategic-safety score through stronger robustness, much lower collusion, and lower externality cost.

3 Experimental Setup

We evaluate four domains chosen to expose different strategic pathologies.

- **Repeated social dilemma:** an iterated prisoner’s-dilemma-style environment.
- **Shared-resource allocation:** repeated extraction from a regenerating common pool.
- **Auction-like competition:** repeated bidding with private values and seller-revenue effects.
- **Language-anchored negotiation:** repeated bargaining over a 10-unit split using natural-language-mapped offer templates.

We compare three main objectives: **reward-only**, **welfare-only**, and **SEPO**. For ablation, we additionally remove one safety term at a time: **SEPO-no-exploit**, **SEPO-no-collusion**, and **SEPO-no-externality**. The optimizer is the same cross-entropy search in all cases.

The primary metrics are individual payoff, social welfare, exploitability, robustness under opponent shift, collusion risk, externality cost, and a strategic-safety index

$$S(\pi) = u(\pi) - 3e(\pi) - 4c(\pi) - 2x(\pi),$$

which is reported only as a summary score. The component metrics remain the main evidence.

4 Main Results

Table 1 reports the aggregate results across the four domains. Reward-only optimization delivers the highest raw payoff, but it also produces the highest collusion risk and the highest externality cost. SEPO gives up 0.78 payoff points relative to reward-only, but it improves opponent-shift robustness from 12.94 to 13.51, reduces collusion risk from 0.250 to 0.016, lowers externality cost from 0.643 to 0.486, and achieves the best strategic-safety index overall.

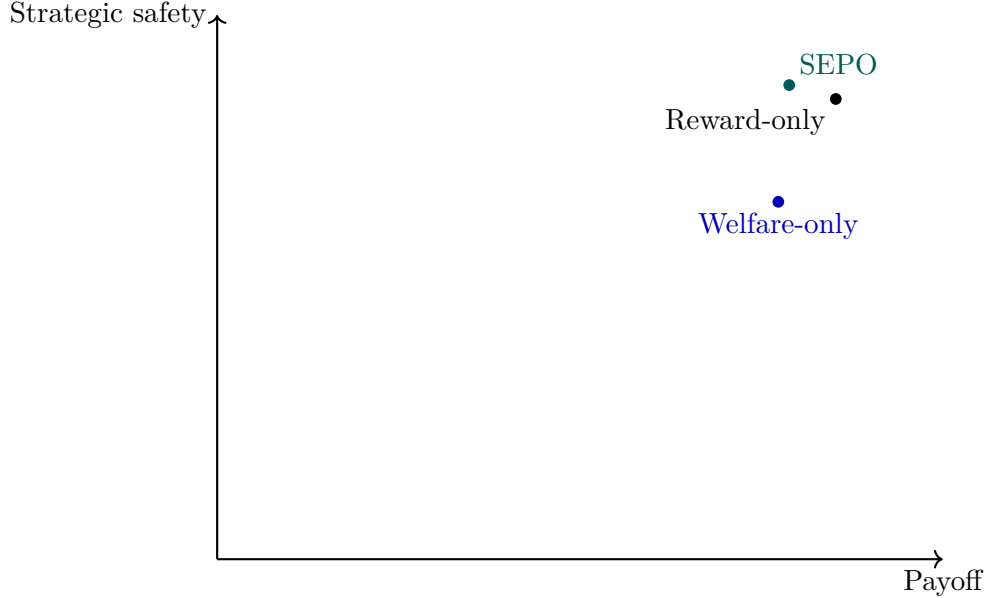


Figure 2: Raw payoff versus strategic safety. SEPO shifts the frontier toward safer strategic behavior with only a moderate reduction in private payoff.

The main tradeoff is clear. Reward-only policies are locally strong but systemically aggressive. Welfare-only policies improve aggregate welfare but are highly exploitable and strategically weak. SEPO lands in the middle: it is not the highest-payoff objective, but it is the most balanced one. This is the pattern the paper was designed to test.

4.1 Domain Breakdown

Table 2 shows where the gains come from. The clearest win is shared-resource allocation. Reward-only converges almost entirely to **high-extract**, which yields high individual reward but maximal collusion risk and severe environmental depletion. SEPO moves instead toward **medium-extract** and **scarcity-aware** play, lowering collusion risk from 1.0 to 0.066 and reducing externality cost from 1.486 to 0.855. That single domain explains most of the aggregate safety gain.

In auctions and negotiation, all methods converge to relatively conservative policies, so the differences are smaller. Social dilemmas remain challenging: SEPO prefers a grim-trigger-like policy that preserves strong payoff but does not substantially reduce exploitability. This is an important negative result, because it shows that the current objective helps most where harmful coordination and environmental cost dominate, and less where exploitability is driven by hard strategic reciprocity.

5 Ablations and Diagnostics

Table 3 isolates the contribution of each safety term. Removing the collusion penalty causes the largest collapse back toward reward-only behavior: payoff rises, but collusion risk and externality both return to unsafe levels. Removing the externality term also degrades the safety index. Removing the exploitability term matters less in the current domains, which is consistent with the domain analysis above: SEPO’s strongest empirical benefit is in preventing harmful coordination

Domain	Objective	Payoff $\uparrow$	Exploit. $\downarrow$	Collusion $\downarrow$	Externality $\downarrow$
Social dilemma	Reward-only	18.333	0.208	0.000	0.202
	Welfare-only	16.058	1.630	0.000	0.059
	SEPO	18.283	0.208	0.000	0.204
Resource	Reward-only	11.333	0.000	1.000	1.486
	Welfare-only	9.683	0.457	0.853	0.602
	SEPO	8.225	0.326	0.066	0.855
Auction	Reward-only	4.688	0.237	0.000	0.215
	Welfare-only	4.700	0.237	0.000	0.216
	SEPO	4.654	0.242	0.000	0.219
Negotiation	Reward-only	6.583	0.000	0.000	0.671
	Welfare-only	6.667	0.000	0.000	0.667
	SEPO	6.667	0.000	0.000	0.667

Table 2: Domain-level results. Shared-resource allocation is the clearest case where reward-only optimization is strategically unsafe and SEPO changes the selected policy family.

Objective	Payoff $\uparrow$	Exploit. $\downarrow$	Collusion $\downarrow$	Safety $\uparrow$
Reward-only	10.118	0.118	0.251	7.468
Welfare-only	9.186	0.576	0.213	5.835
SEPO	9.367	0.210	0.007	7.757
SEPO-no-exploit	9.304	0.259	0.011	7.686
SEPO-no-collusion	10.169	0.126	0.251	7.506
SEPO-no-externality	10.138	0.133	0.238	7.531

Table 3: Ablations of the SEPO penalties. The collusion and externality terms are the main drivers of the aggregate safety improvement.

and environmental damage, not in fully solving exploitability.

6 Related Work

SEPO sits at the intersection of multi-agent reinforcement learning, game-theoretic equilibrium analysis, and safety-oriented agent optimization. The closest conceptual line is work on exploitability minimization and equilibrium-aware training in games, but that literature usually emphasizes competitive robustness rather than explicit social safety penalties. A second line studies cooperative and socially aware optimization, where the goal is higher welfare or reduced conflict. SEPO differs from both by explicitly optimizing a three-way tradeoff among exploitability, collusion risk, and externality cost.

The paper is also related to emerging work on language agents in strategic environments. Our implementation is intentionally lightweight: it uses language-anchored policy proxies and interpretable strategy libraries instead of full-scale LM fine-tuning. That makes the current contribution an objective-design study rather than a deployment-scale language-model paper. The key question is not whether these exact strategy libraries are realistic. It is whether an equilibrium-aware objective selects better strategic behavior than reward-only optimization under a controlled comparison. The answer, in the current study, is yes.

7 Limitations

This paper does not fine-tune a frontier language model directly. The experiments use controlled multi-agent proxy environments with interpretable strategy libraries, including one language-anchored negotiation domain. That keeps the causal question clean, but it limits ecological realism.

The exploitability results are mixed: SEPO substantially improves robustness, collusion, and externalities, but it does not dominate reward-only on every domain-specific exploitability proxy. The next step is to port the same objective to actual LM-agent fine-tuning and evaluate on richer multi-agent benchmarks with natural-language state and action spaces.

8 Conclusion

Reward-only optimization is too weak an objective for strategic agents. It selects policies that can perform well on private payoff while remaining collusive, environmentally harmful, or brittle under partner shift. SEPO addresses that gap by adding explicit equilibrium-aware safety terms to the objective. In controlled multi-agent domains, the resulting policies give up some raw payoff but achieve stronger system-level behavior and the best aggregate strategic-safety score. The main empirical lesson is practical: if we want strategically aligned agents, we have to optimize for more than reward.

References

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A Appendix: Experimental Details

Strategy libraries. The four domains use finite strategy libraries rather than unconstrained policy classes. Social dilemmas include always-cooperate, always-defect, tit-for-tat, generous tit-for-tat, and grim trigger. Shared-resource allocation includes low, medium, and high extraction, plus scarcity-aware and reciprocal extraction. Auction-like competition includes truthful, aggressive, conservative, shaded, and collusive-low bidding. Negotiation includes fair, balanced, selfish, hardball, and concede variants.

Optimization. Each objective is optimized by cross-entropy search over mixture weights on the strategy library. Warm starts are objective-specific priors derived from solver- and simulation-inspired heuristics. Reward-only, welfare-only, and SEPO use the same optimization budget. The ablations remove one penalty term at a time while keeping the optimizer fixed.

Reproducibility. The reproducibility pack includes the full experiment runner, aggregate metrics, domain-level results, ablation tables, trajectory examples, and the exact run command used to generate the reported outputs.